Annals of Operations Research (2020) 292:47–70
https://doi.org/10.1007/s10479-020-03654-0

ORIGINAL RESEARCH

Sustainable, multimodal and reliable supply chain design

Ozgur Kabadurmus[1] · **Mehmet S. Erdogan**[1]

Published online: 6 June 2020
© Springer Science+Business Media, LLC, part of Springer Nature 2020

Abstract
Emerging issues and new challenges of globalization have forced companies to design their supply chains for not only minimizing cost but also considering other factors. Supply chains are exposed to new environmental regulations to reduce their carbon emissions and compelled to consider other overlooked factors, such as risk. In this paper, we consider a multi-echelon multimodal supply chain network design problem with multiple products and components that take economic, environmental and risk factors into account. The problem is modeled as a Mixed Integer Linear Programming model and constrained by a carbon cap-and-trade scheme and a risk threshold. This novel problem realistically portrays the supply chain network design considering sustainability and reliability factors simultaneously. The proposed model has been tested on randomly generated hypothetical but realistic test instances. The impacts of different risk thresholds and unit carbon prices on the supply chain cost, risk and emissions are analyzed. The effects of multimodal transportation modes on cost, risk and emissions are also tested. Results prove that using multimodal transportation decreases supply chain cost and carbon emission. In addition, the total supply chain cost and carbon emission increase if the decision maker is risk-averse. The choice of transportation modes is sensitive only to emission levels.

Keywords Supply chain design · Sustainability · Supply chain risk · Multimodal transportation · Mixed integer linear programming

1 Introduction

Supply chains include various players such as suppliers, distributors, wholesalers, retailers and customers. These players interact with each other incessantly and any decision made by one of them affects the others in the chain. The existence of this complex web of interactions means that any unforeseen event or any unconsidered parameter within the chain makes a supply chain diverge from its optimal operational conditions. Therefore, this divergence may cause inefficiencies in the supply chain.

Inefficiencies in a supply chain make the supply chain operate at a higher cost. Therefore, the minimization of supply chain cost has been the main focus of the supply chain optimization

✉ Ozgur Kabadurmus
ozgur.kabadurmus@yasar.edu.tr

[1] Department of International Logistics Management, Yasar University, 35100 Bornova, Izmir, Turkey

problems. Previous studies mostly considered more apparent costs, such as raw material, transportation, ordering and production cost. However, some overlooked or neglected factors may cause direct or indirect costs for companies. Carbon emissions, supply chain risk and multimodal transportation can be considered as these factors.

One of the most popular topics of the last decade is global warming which affects the entire world. And, global warming has been worsening steadily. To alleviate the effects of global warming, new treaties were signed and new regulations were put into effect by governments on international and regional levels such as Kyoto Protocol, Paris Agreement, and Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA). Consequently, these treaties and regulations have pressured numerous companies since their supply chain operations play a significant role in global warming. The most important governmental regulations related to global warming are the concepts of the carbon tax and carbon cap-and-trade.

Applying carbon tax to companies exceeding their pre-determined total carbon emission limit (carbon cap) has become widespread after the pioneering countries such as Australia, New Zealand, Canada and China (Goulder and Schein 2013). The unit carbon price is determined externally by the regulations in the carbon tax scheme. And, the regulatory authorities reduce carbon cap yearly to force companies to be "greener" (Avi-Yonah and Uhlmann 2009). Although environmentally friendly policies can be cost-saving (Zissis et al. 2018) in certain scenarios, carbon cap limitation causes under-utilization of resources and lost sales to companies since they are reluctant to pay carbon taxes. However, this created carbon trade opportunities for some companies. Carbon trade is simply the process of selling or buying the allowance of carbon dioxide emissions among companies. Under a pure carbon cap-and-trade scheme, the unit carbon price is determined by the market. At the end of each year, a company must use enough allowance to cover its emissions, otherwise, a significant tax is imposed. Therefore, companies trade their carbon emission allowances with others as needed. In this paper, we included this carbon cap-and-trade system in our model.

There are various risk factors at each stage of a supply chain. Failures to make deliveries, inability to meet quality requirements, a sudden increase in cost, transit time variability, lack of supplier visibility, lack of integration with suppliers, low technical reliability and supplier bankruptcy are considered as some of the risk factors related to the suppliers (Wu et al. 2006; Kull and Talluri 2008; Zsidisin and Ellram 2003; Chopra and Sodhi 2004). Employee accidents, operator absences, lack of experience or training, product quality and safety, the rate of product obsolescence, insufficient maintenance, unstable manufacturing processes, design changes and technological advancements can be viewed as manufacturing process risks (Tummala and Schoenherr 2011; Tuncel and Alpan 2010; Wu et al. 2006; Ho et al. 2015). The risk of a supplier can be assessed using the past data and the risk values of all suppliers can be found. This risk assessment plays a significant role in the supply chain's efficiency by helping the decision maker to make better decisions to keep the supply chain under a certain risk level.

Multimodality is considered as another important trend in global (and also local) transportation. Multimodal transportation is the utilization of different transportation modes during the shipment of cargo, whereas unimodal transportation uses only a single mode of transportation. With the increased global trade operations, multimodal transportation provides green transportation options to logistics companies. Also, multimodal transportation reduces costs, however, its effects on risk levels have not been investigated yet.

As the intense competition for global supply chain management and logistics operations increases, a growing number of companies try to pursue operational excellence in their supply chains so that they gain a better market share for sustainable development. The supply chain designs that overlook sustainability, risk or multimodality may not be competitive in the long

run as the intense global competition and highly demanding customers require these factors to be included. Therefore, to grasp the dynamics of the global supply chains and increase competitiveness, the consideration of sustainability, reliability and multimodality in supply chains is essential.

However, most studies on supply chain design have only focused on minimizing cost. Even though some studies tackle the sustainable and multimodal supply chain design problem (Liotta et al. 2015; Rezaee et al. 2015; Le and Lee 2013) or supply chain reliability (Ahranjani et al. 2020; Kaur and Singh 2019; Fahimnia et al. 2018; Rahmani and Mahoodian 2017) individually, to the best of our knowledge, the issue of supply chain reliability has not been considered together with sustainability or multimodality in the supply chain design literature. The previous supply chain reliability studies mainly considered disruptions and risk in supply chains, however, we included the supplier reliability to realistically portray the supply chain network design. As for reducing carbon emissions, 40 national and 25 local jurisdictions implemented or scheduled cap-and-trade programs until the end of 2017 (World Bank Group 2017). Since many countries have attempted to enact legislation or design a market-based carbon trading mechanism, our study also employs a carbon cap-and-trade scheme to reduce carbon emissions.

This article presents a novel multi-commodity three-echelon supply chain optimization model considering not only carbon cap-and-trade but also risk factors of suppliers. The model also allows multimodal transportation operations. As transportation and manufacturing are two major emission sources in supply chains (Waltho et al. 2019), carbon emissions due to transportation and production operations are considered in our study. Also, the total risk of the supply chain operations is kept below a predetermined "risk threshold". The proposed model has been tested on a hypothetical but realistic case study and the effects of carbon cap-and-trade policy, risk and multimodal transportation on supply chain cost, risk and emission are discussed. Since carbon cap-and-trade is found to be an effective tool if the cap and the price are selected carefully (Waltho et al. 2019), we also performed an extensive sensitivity analysis on emission price, risk threshold and multimodality.

The remainder of the article is organized as follows. Section 2 summarizes the prior relevant studies. Section 3 presents the problem definition and the mathematical model. Section 4 summarizes the computational results. Conclusions and future work are given in Sect. 5.

2 Literature review

2.1 Sustainability in supply chains

Environmental sustainability in supply chains became an important issue in the last two decades. Different approaches have been applied in the literature to design sustainable supply chains.

The majority of the studies added carbon emission as a cost factor to the objective function (Diabat and Al-Salem 2015; Choudhary et al. 2015; Fahimnia et al. 2015; Le and Lee 2013; Liotta et al. 2015; Alizadeh et al. 2019; Zhang et al. 2017; Shaw et al. 2016). In most studies, deterministic demand is considered (Choudhary et al. 2015; Fahimnia et al. 2015; Le and Lee 2013; Liotta et al. 2015; Alizadeh et al. 2019; Zhang et al. 2017; Shaw et al. 2016), however, Rezaee et al. (2017) and Han et al. (2017) assumed stochastic parameters (e.g., demand).

While Le and Lee (2013), Alizadeh et al. (2019) and Fahimnia et al. (2015) considered multiple products types and multiple periods, Liotta et al. (2015) and Rezaee et al. (2017) assumed a single-period model with multiple product types. Single period and single product type are considered by Diabat and Al-Salem (2015), Zhang et al. (2017), Shaw et al. (2016) and Choudhary et al. (2015). Similar to our study, Fahimnia et al. (2015) and Liotta et al. (2015) extended multiple product types by adding component requirements for each product. Raw material requirements of each component were also included by Le and Lee (2013).

Similar to our study, multiple production facilities with activation cost (i.e., the cost of operating a facility) were considered in the majority of the studies (Diabat and Al-Salem 2015; Choudhary et al. 2015; Fahimnia et al. 2015; Rezaee et al. 2017; Liotta et al. 2015; Rabbani et al. 2018; Alizadeh et al. 2019; Zhang et al. 2017; Shaw et al. 2016). Warehouses were only included in Choudhary et al. (2015), Rezaee et al. (2017), Liotta et al. (2015), Shaw et al. (2016), Zhang et al. (2017) and Le and Lee (2013). In addition, Rabbani et al. (2018) assumed that the capacities of the plants can be expanded with an extra cost. While the models of Diabat and Al-Salem (2015) and Choudhary et al. (2015) have no suppliers, Rezaee et al. (2017), Liotta et al. (2015), Fahimnia et al. (2015), Alizadeh et al. (2019), Zhang et al. (2017), Shaw et al. (2016) and Le and Lee (2013) considered multiple suppliers. The availability of products in suppliers changes over a set of scenarios in the study of Alizadeh et al. (2019). As a distinction according to the transportation modes, Choudhary et al. (2015), Fahimnia et al. (2015) and Diabat and Al-Salem (2015) considered unimodal transportation, whereas Liotta et al. (2015), Rezaee et al. (2017), Alizadeh et al. (2019), Zhang et al. (2017) and Le and Lee (2013) allowed multimodal transportation. In our study, we also allow multimodal transportation in our model. Similar to our model, Choudhary et al. (2015) considered a carbon cap-and-trade scheme where the amount of carbon emission under (above) the carbon cap will be sold (bought). Rezaee et al. (2017) also included carbon cap constraints and investigated different carbon price scenarios. In our analysis, we also investigated the effects of different carbon prices on supply chain performance. However, unlike our study, none of these sustainable supply chain studies considered reliability.

2.2 Sustainability and reliability in supply chains

Various studies investigated the dynamics of sustainable and reliable supply chains. However, as summarized in a recent review (Hosseini et al. 2019), these studies considered reliability mainly as supply chain disruptions. For example, Rahmani and Mahoodian (2017) designed a multi-product three-echelon sustainable and reliable supply chain network. Their study considered uncertainty risk and facility disruption risk. Uncertainty risk includes the risk of uncertain demand and cost of manufacturing. To model the facility disruption risk, they considered various scenarios that exclude some of the factories and distribution centers as disruptions. In a similar study, Fahimnia et al. (2018) proposed a three-echelon supply chain where manufacturing plants and distribution centers are subject to disruptions. A set of scenarios were generated to represent different disruption possibilities and an elastic p-robust constraint included for scenarios to guarantee that the supply chain is reliable to a certain level. To include sustainability, they ensured that the environmental performance of each operation (production, storage, and inbound and outbound transportation) is greater than a threshold. Kaur and Singh (2019) formulated a sustainable multi-period multi-product procurement and logistics for disaster resilient supply chain where suppliers and carriers have varying capacities to cope with disasters. Similar to previous studies, they tested the resilience of their model using randomly generated disaster scenarios. Ahranjani et al. (2020) formulated a

multi-period multi-product biofuel supply chain. They considered possible drought scenarios causing disruption of biomass feedstock and used robust optimization to minimize the total expected cost while limiting greenhouse gas emissions. To include sustainability, they limited the emissions in their mathematical model. None of the sustainable and reliable supply chain studies considered supplier reliability as in our study and their reliability representations are quite different than ours. In our study, instead of using disruption scenarios, we modeled supplier reliabilities to maximize supply reliability.

2.3 Multimodality in supply chains

As stated by Waltho et al. (2019), multimodal studies are not very common in supply chain network design literature. Similar to our study, multimodal supply chain design studies in the literature consider two or more transportation modes in their models, however, our study allows all transportation modes (road, rail, air and water) in the network design problem. Liotta et al. (2015) and Zakeri et al. (2015) allowed road, rail and water transportation options among the arcs of their three-echelon supply chain structures. Chaabane et al. (2011), Shaw et al. (2016) and Ramudhin et al. (2008) employed rail, air, and road transportation modes in their three-echelon supply chain models and Shaw et al. (2016) concluded that using a combination of transportation modes is highly effective to decrease carbon emission. In addition, Moradinasab et al. (2018) assumed road, rail and pipeline transportation modes in a petroleum green supply chain. Different than the other studies, Le and Lee (2013) also considered transportation between distributors, however, they allowed multimodal transportation only among some specific arcs. Similarly, Fahimnia et al. (2018) allowed different transportation mode options for each arc. Alizadeh et al. (2019) considered multimodal facilities where trucks arrive and the products are transferred to water or rail transportation modes. Similar to our study, transportation capacities are determined according to volume and weight limits of the vehicles in the models of Ramudhin et al. (2008) and Chaabane et al. (2011), whereas the capacities are defined in terms of the number of units in the rest of the studies. Unlike our study, none of these studies addressed supply chain reliability and sustainability issues with a carbon cap-and-trade scheme.

3 Problem definition

The proposed supply chain model herein composes of multiple suppliers, plants, customers and products. Our model also includes the bill of materials (BOM) requirements for each product where BOM defines the information regarding final products and their components (Trappey et al. 1996). If a component is required for a final product, only one unit of the component is assumed to be used. Each final product requires a set of components that are bought from suppliers. After the required components for a final product arrive at a manufacturing plant, the product is produced at that plant. Moreover, the total cycle time of a plant (i.e., the production times of all products that are produced in that plant) cannot exceed a predetermined cycle time limit. The production time of a final product is defined as the time to finish all operations of that product (Liotta et al. 2015). Upon the completion of the production process, products are delivered directly to the end customers. Shortages are not allowed and inventory holding costs are not included in our model.

Multimodal transportation is also considered in our model. Products can be transported from one echelon to another via road, rail, water or air transportation modes. There are two

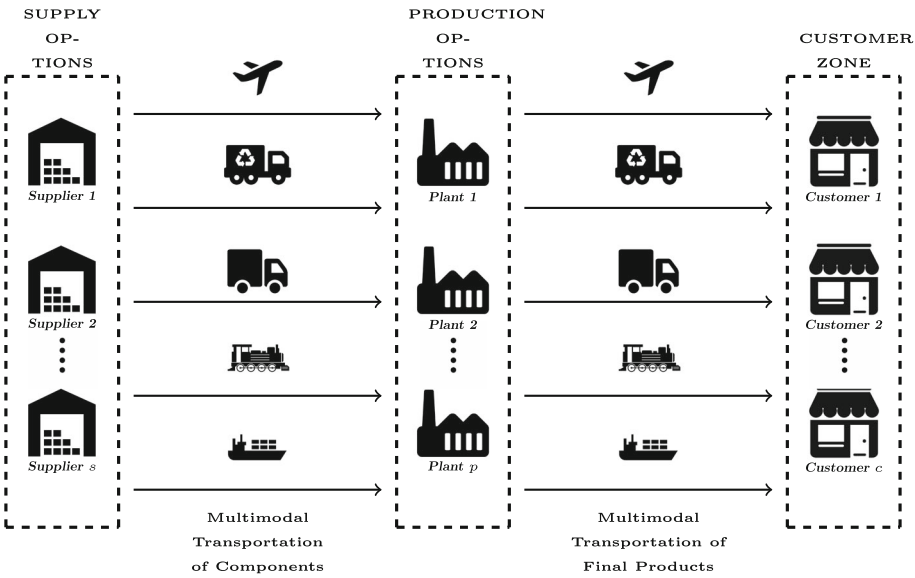


Fig. 1 The supply chain structure of the proposed model

different truck types in road transportation: regular and environment-friendly trucks. While regular trucks offer a lower cost to the company, environment-friendly trucks emit less carbon dioxide. To be more realistic, rail, water and air transportation are allowed only in some arcs with long distances, while road transportation is allowed in all arcs. Figure 1 illustrates the supply chain structure of our model.

Transportation cost is calculated per vehicle (container) and per kilometer for road, rail and water transportation modes. Note that Full Truck Load (FTL) transportation is assumed for road transportation mode, and similarly, Full Container Load (FCL) transportation is assumed for water and rail transportation modes. Therefore, the transportation costs of road, rail and water modes are proportional to the distance and the number of vehicles (containers). However, to realistically capture the pricing practice of the air freight industry, the cost of air transportation is calculated by multiplying the arc distance and the billable weight of the consignment that is sent over an arc. The billable weight is the maximum of the physical weight and the volumetric weight of cargo. Volumetric weight is calculated by dividing the volume (m^3) of the cargo by 0.006. Although some companies (e.g., UPS and DHL) recently started to use 0.005 instead of 0.006, we chose 0.006 as it is used by USPS (United States Postal Services). Similar to the road, rail and water transportation modes, to calculate the emission of air transportation, product weight is used instead of volumetric weight.

Furthermore, each supplier has a risk value between 0 and 1. The risk of a supplier is determined according to many factors, such as quality issues, delivery speed, and billing issues. Each component has an importance level between 1 and 10 to change its impact on the total risk calculation where the total risk is simply the weighted sum of risks in terms of the importance levels. Our model keeps the total risk below a predefined risk threshold. As the total risk level decreases, the supply chain becomes more reliable. To focus only on supply risk, the risks of the plants are not considered in our model. In addition, our model assumes that a company buys (sells) carbon allowance equals to the quantity of emission over (under) the carbon cap.

3.1 Mathematical model

The mixed integer linear programming (MILP) model of the problem is given below.

Sets

A	Arcs, $(i, j) \in A : i, j \in S \cup M \cup C$
B	Component requirements of final products, $(k, f) : k \in K, f \in F$
C	Customers
F	Final products
K	Components
M	Manufacturing plants
P	All products: $K \cup F$
S	Suppliers
T	All transportation modes
$\bar{T}$	Air transportation mode
$T^{'}$	All transportation modes except air transportation, $T^{'} = T \backslash \bar{T}$

Parameters

a_i	Activation cost of supplier/plant $i \in S \cup M$
c_t	$\begin{cases}\text{Variable transportation cost per km per vehicle (container) for mode } t \in T^{'} \\ \text{Variable transportation cost per km per billable weight (kg) for mode } t \in \bar{T}\end{cases}$
d_{ij}	Distance (km) of arc $(i, j) \in A$
q_{jf}	Demand of customer $j \in C$ for final product $f \in F$
e_{if}	Carbon emission (kg) of producing one unit of final product $f \in F$ at plant $i \in M$
$\bar{e}_t$	Carbon emission (kg) per kilometer per weight (metric ton) for transportation mode $t \in T$
f_t	Fix cost of using one unit of vehicle/container for transportation mode $t \in T^{'}$
l_{ip}	Capacity (number of products) of supplier/plant $i \in S \cup M$ to produce $p \in P$
l^v_{ijt}	Total vehicle/container capacity of arc $(i, j) \in A$ for transportation mode $t \in T$
p_{ip}	Purchasing (production) cost of supplier (plant) $i \in S \cup M$ for one unit of component (product) $p \in P$
r_i	Risk value (between 0 and 1) of supplier $i \in S$
v_p	Volume (m^3) of product $p \in P$
v^{cap}_t	Total volume (m^3) capacity of a vehicle/container/airplane of $t \in T$
w^b_p	Billable weight (kg) of product $p \in P$
w_p	Weight (kg) of product $p \in P$
w^{cap}_t	Total weight (kg) capacity of a vehicle/container/airplane of $t \in T$
α	Market price of one unit (kg) carbon emission allowance
β_k	The importance level (a reel number value between 1 and 10) of component $k \in K$
τ_{if}	Production time (min) of final product $f \in F$ at plant $i \in M$
τ^c_i	Maximum cycle time (min) of plant $i \in M$
φ	Emission limit (kg)
ω	Risk threshold

Decision variables

Q^b	The amount of emission bought from other companies
Q^s	The amount of emission sold to other companies
X_{ijtp}	The number of units of product $p \in P$ transported through arc $(i, j) \in A$ with transportation mode $t \in T$

Y_{ijt} The number of vehicles/containers/airplanes of $t \in T$ used on arc $(i, j) \in A$

Z_{ip} $\begin{cases} 1 & \text{if supplier/plant } i \in S \cup M \text{ is activated for product } p \in P, \\ 0 & \text{otherwise} \end{cases}$

Objective function

$$\begin{aligned} z = & \sum_{(i,j)\in A} \sum_{t\in T'} c_t * Y_{ijt} * d_{ij} + \sum_{(i,j)\in A} \sum_{t\in \bar{T}} \sum_{p\in P} w_p^b * X_{ijtp} * c_t * d_{ij} \\ & + \sum_{(i,j)\in A} \sum_{t\in T'} Y_{ijt} * f_t + \sum_{(i,j)\in A} \sum_{t\in T} \sum_{p\in P} X_{ijtp} * p_{ip} \\ & + \sum_{i\in S\cup M} \sum_{p\in P} Z_{ip} * a_i + \alpha * (Q^b - Q^s) \end{aligned} \tag{1}$$

Subject to

$$\sum_{i\in M} \sum_{t\in T} X_{ijtf} = q_{jf} \quad \forall f \in F, j \in C \tag{2}$$

$$\sum_{i\in S} \sum_{t\in T} X_{ijtk} = \sum_{(k,f)\in B} \sum_{i\in C} \sum_{t\in T} X_{jitf} \quad \forall k \in K, j \in M \tag{3}$$

$$\sum_{j\in C} \sum_{t\in T} \sum_{f\in F} \tau_{if} * X_{ijtf} \le \tau_i^c \quad \forall i \in M \tag{4}$$

$$\sum_{j\in M\cup C} \sum_{t\in T} X_{ijtp} \le l_{ip} * Z_{ip} \quad \forall p \in P, i \in S \cup M \tag{5}$$

$$Y_{ijt} \le l_{ijt}^v \quad \forall (i, j) \in A, t \in T \tag{6}$$

$$\sum_{p\in P} X_{ijtp} * v_p \le v_t^{cap} * Y_{ijt} \quad \forall (i, j) \in A, t \in T \tag{7}$$

$$\sum_{p\in P} X_{ijtp} * w_p \le w_t^{cap} * Y_{ijt} \quad \forall (i, j) \in A, t \in T \tag{8}$$

$$w_p^b = \max\left(w_p, \frac{v_p}{0.006}\right) \quad \forall p \in P \tag{9}$$

$$\begin{aligned} & \sum_{(i,j)\in A} \sum_{t\in T} \sum_{p\in P} \frac{w_p}{1000} * X_{ijtp} * \bar{e}_t * d_{ij} \\ & + \sum_{i\in M} \sum_{j\in C} \sum_{t\in T} \sum_{f\in F} X_{ijtf} * e_{if} \le \varphi + Q^b - Q^s \end{aligned} \tag{10}$$

$$\frac{\sum_{i\in S} \sum_{j\in M} \sum_{t\in T} \sum_{k\in K} X_{ijtk} * r_i * \beta_k}{\sum_{(k,f)\in B} \sum_{j\in C} q_{jf} * \beta_k} \le \omega \tag{11}$$

$$X_{ijtk}, Y_{ijt} \in \{0, 1, 2, ...\}; \; Z_{if} \in \{0, 1\}; \text{ and } Q^b, Q^s \ge 0 \tag{12}$$

Objective function (1) minimizes the total supply chain cost. The first and second parts of the objective function include variable transportation costs. Note that variable costs of road, rail and water modes are calculated per vehicle per kilometer, whereas the variable cost of air transportation is calculated per billable weight per kilometer. The third part adds the fix costs of road, rail and water transportation modes. The fourth and fifth parts of the objective function consider purchasing (production) cost and supplier (plant) activation cost, respectively. The sixth part adds the cost (or profit) of emission trade. Constraint (2) ensures

that customer demand is satisfied. Constraint (3) guarantees that the required components to produce final products are transported to each plant. Constraint (4) prevents the total cycle time of products from exceeding the cycle time limit of the plant. Constraint (5) is the capacity limitation of suppliers (plants) for each component (finished products). Constraint (6) guarantees that the number of activated trucks/containers/airplanes does not exceed the capacity. Constraint (7) ensures that the total volume of products transported via a transportation mode on an arc does not exceed the total volume capacity of that arc due to the number of activated trucks/containers/airplanes. Similar to Constraint (7), Constraint (8) limits the capacity of an arc for a transportation mode in terms of total weight. Constraint (9) calculates the billable weight of each product by taking the maximum of the product's physical weight and its volumetric weight. Constraint (10) ensures that the total amount of carbon dioxide emission plus the emission trade (the amount bought or sold) is equal to the emission limit. Constraint (11) prevents the total supply chain risk from exceeding the risk threshold. Boundary constraints on the decision variables are defined in Constraint (12).

4 Results

4.1 Data generation

The size of the problem is determined by the numbers of suppliers, plants, customers, products and components. It is also affected by the number of transportation options. The proposed supply chain model is applicable for any supply chain network type from regional to global or small to large as the numbers of suppliers, plants, customers, products, and components can easily be increased and multimodal transportation options can be realistically deployed among them. In this study, we assumed a global supply chain for our case study. To analyze our proposed model, we created a hypothetical but realistic supply chain case with three echelons consisting of suppliers, plants and customers. Customer demand, capacity, cost, distance, weight/volume of products, emission and risk data are generated randomly as explained in Table 1. Data regarding supplier and plant capacities are generated in such a way that all of the component (product) demand cannot be fulfilled by a single supplier (plant) while the total capacity of all suppliers (plants) is sufficient to fulfill the total demand as in most supply chains in real life. Similarly, the production times of products are generated in such a way that all products cannot be produced by a single plant while the total working time of all plants is enough to produce all products. Since we assumed that a product requires three components, components must have less cost, weight and volume than products. Therefore, purchasing cost, weight and volume of components are generated proportionally to production cost, weight, and volume of products. The range of carbon emission for producing a final product is chosen according to Shaw et al. (2013). However, we slightly reduced their values since our model purchases the components from suppliers to manufacture and assemble the products in plants. Risk values of suppliers are generated realistically so that each supplier has both low and high risk components, and similarly, each component has both low and high risk suppliers.

Maximum allowed weight and volume, variable cost, activation cost and unit emission are fixed parameters related to the transportation modes. Table 2 shows these parameter values. In our study, carbon emissions of water and rail transportation are obtained from McKinnon and Piecyk (2010), while carbon emissions of road and air transportation are obtained from Piecyk et al. (2015). Weight and volume capacities are defined according to container type

Table 1 Data generation of the case study

Parameter	Data generation
Average demand of a customer	Uniform random number between 400 and 600
Demand variance of a customer	Uniform random number between 1/4 and 1/3 of the average demand
Actual demand of a customer	Normal distribution according to the average demand and the variance
Plant capacity for a final product	Uniform random number between c and 1.5*c, where $c = \frac{\sum_{j \in C} q_{jf}}{\lvert M \rvert}, \forall f \in F$
Supplier capacity for a component	Uniform random number between s and 1.5*s, where $s = \frac{\sum_{j \in C} \sum_{(k,f) \in B} q_{jf}}{\lvert S \rvert}, \forall k \in K$
Production cost of a final product	Normal distribution with the mean m and the standard deviation m*0.05, where m is a uniform random number between 10 and 100
Purchasing cost of a component	Normal distribution with the mean r and the standard deviation r*0.05, where r is a uniform random number between 1 and 10
Carbon emission (in kg) for producing a final product	Uniform random number number between 0.02 and 0.08
Risk of a supplier	Uniform random number between 0.05 and 0.20
Production time of a final product in a plant	Uniform random number between t and 1.5*t, where $t = \frac{\tau_i^c}{\sum_{j \in C} \sum_{f \in F} q_{jf} / (\lvert M \rvert * \lvert F \rvert)} / 2, \forall f \in F$
Location of a supplier/plant/customer	Each node is randomly located on a square grid of size 4000
Distance of arc (i, j)	Euclidean distance is calculated between i and j
Weights (kg) of a final product and a component	The weight of a final product is a uniform random number between 10 and 25, and the weight of a component is 5
Volumes (m^3) of a final product and a component	The volume of a final product is a uniform random number between 0.3 and 1, and the volume of a component is 0.15

for road, rail, and water transportation modes. A 45-foot container is considered for road transportation mode while a 40-foot container is considered for rail and water transportation modes. As for air transportation, a Boeing 777-200 aircraft is considered. The maximum cycle time of all plants is assumed to be 10,560 min to reflect the monthly working time (22 days per month and 8 h per day). The activation cost of a plant/supplier is taken as 2000.

Before the determination of the number of available vehicles (or containers) of each vehicle type, the number of vehicles (or containers) required to transport the average number of products (components) in each transportation mode t, denoted as b_t, is calculated as shown in Eqs. (13) and (14). Note that each arc is assumed to have enough capacity to fulfill the total demand using a sufficient number of vehicles (or containers).

For the arcs between plants and customers:

$$b_t = \max \left\{ \frac{\sum_{j \in C} \sum_{f \in F} q_{jf} * w_f}{w_t^{cap} * |M| * |C|}, \frac{\sum_{j \in C} \sum_{f \in F} q_{jf} * v_f}{v_t^{cap} * |M| * |C|} \right\} \tag{13}$$

Table 2 Parameter values of the transportation modes

	Trucks		Rail	Water	Air
	Regular	Environmental			
a_i (per vehicle)	150	180	170	150	0
c_t (per km per vehicle)	0.075	0.085	0.035	0.05	0.0025[a]
$\bar{e}_t$ (per km per kg)	0.075	0.062	0.025	0.015	0.6
v_t^{cap} (m^3)	90	90	67	67	60
w_t^{cap} (kg)	24,000	24,000	26,000	26,000	21,000

[a]The variable cost of air mode is calculated per km per kg

Table 3 Multimodal arc construction rules according to transportation modes

	Trucks		Rail	Water	Air
	Regular	Environmental			
Minimum distance (>km)	0	0	300	1000	800
Probability (%)	100	100	75	25	100

Table 4 The number of vehicles assigned to an arc for each transportation case

Case	Number of vehicles/containers/airplanes				
	Trucks		Rail	Water	Air
	Regular	Environmental			
Unimodal	$\lceil b_t \rceil$	–	–	–	–
Multimodal 1	$\lceil 0.5b_t \rceil$	$\lceil 0.5b_t \rceil$	–	–	–
Multimodal 2	$\lceil 0.5b_t \rceil$	$\lceil 0.5b_t \rceil$	–	–	1
Multimodal 3	$\lceil 0.4b_t \rceil$	$\lceil 0.4b_t \rceil$	$\lceil 0.2b_t \rceil$	–	–
Multimodal 4	$\lceil 0.4b_t \rceil$	$\lceil 0.4b_t \rceil$	$\lceil 0.2b_t \rceil$	–	1
Multimodal 5	$\lceil 0.4b_t \rceil$	$\lceil 0.4b_t \rceil$	–	$\lceil 0.2b_t \rceil$	1
Multimodal 6	$\lceil 0.35b_t \rceil$	$\lceil 0.35b_t \rceil$	$\lceil 0.2b_t \rceil$	$\lceil 0.1b_t \rceil$	1

For the arcs between suppliers and plants:

$$b_t = \max \left\{ \frac{\sum_{j \in C} \sum_{(k,f) \in B} q_{jf} * w_k}{w_t^{cap} * |S| * |M|}, \frac{\sum_{j \in C} \sum_{(k,f) \in B} q_{jf} * v_k}{v_t^{cap} * |S| * |M|} \right\} \tag{14}$$

In the unimodal transportation scenarios, only regular trucks are available. In the multimodal transportation scenarios, the available transportation modes on an arc are determined according to Table 3. On each arc, a transportation mode becomes available according to the arc distance and a probability. Regular and environment-friendly trucks are available on all arcs. Air mode is available on arcs with a distance of 800 km or more. Rail mode is available after 300 km with 75% probability. Water mode is available after 1000 km with 25% probability. According to these rules, six distinct multimodal transportation cases are possible. The rules of assigning the numbers of vehicles, containers or airplanes to an arc for each case are summarized in Table 4.

Table 5 All scenarios

Structure (S–M–C, F–K)[a]	Risk threshold levels	Emission price levels
2–2–2, 3–4	0.08, 0.09, 0.10	0.3, 0.6, 0.9, 1.2
2–2–2, 4–5	0.11, 0.12, 0.13	0.3, 0.6, 0.9, 1.2
3–3–3, 3–4	0.12, 0.13, 0.14	0.3, 0.6, 0.9, 1.2
3–3–3, 4–5	0.11, 0.12, 0.13	0.3, 0.6, 0.9, 1.2
4–4–4, 3–4	0.11, 0.12, 0.13	0.3, 0.6, 0.9, 1.2
4–4–4, 4–5	0.11, 0.12, 0.13	0.3, 0.6, 0.9, 1.2

[a]Suppliers—manufacturing plants—customers, products—components

The emission limit φ is calculated according to Eq. (15). The first part of the emission limit equation calculates the average amount of carbon emission in all plants. The second and third parts of the calculation include the average transportation emissions from suppliers to plants and from plants to customers, respectively. Note that θ is used as the weighted emissions of all transportation modes per kilometer per weight and calculated as in Eq. (16). The indices from 1 to 4 in Eq. (16) represent regular truck, environment-friendly truck, rail and water transportation, respectively and their weights were selected after preliminary experimentation to generate similar emission limits for all instances.

$$\begin{aligned}\varphi = & \sum_{i\in M}\sum_{f\in F} e_{if} * \left(\sum_{j\in C}\frac{q_{jf}}{|M|}\right) \\ & + \theta * \frac{\sum_{j\in C}\sum_{(k,f)\in B} q_{jf} * w_k}{1000} * \frac{\sum_{i\in S}\sum_{j\in M} d_{ij}}{|S| * |M|} \\ & + \theta * \frac{\sum_{j\in C}\sum_{f\in F} q_{jf} * w_f}{1000} * \frac{\sum_{i\in M}\sum_{j\in C} d_{ij}}{|M| * |C|} \end{aligned} \tag{15}$$

$$\theta = 0.25 * \bar{e}_1 + 0.25 * \bar{e}_2 + 0.4 * \bar{e}_3 + 0.1 * \bar{e}_4 \tag{16}$$

To analyze the proposed MILP model, 72 multimodal and 72 unimodal scenarios are generated as summarized in Table 5. Since the MILP model was unable to find a feasible solution for the scenarios with more than 12 nodes, 4 products and 5 components within 1500 s, the scenarios that we generated include a maximum of 12 nodes, 4 products and 5 components. After preliminary experimentation runs of each scenario, three specific risk threshold levels were chosen for each scenario where the impacts of these threshold levels on supply chain performance are the highest. 10 problem instances were randomly generated for each scenario and a total of 1,440 instances were solved with a time limit of 1500 s. The MILP model was coded in IBM CPLEX Optimization Studio and executed on an Intel(R) Core(TM) i7-4510 CPU with 2.60 GHz PC with 8.00 GB memory.

4.2 Findings

To analyze the emission trade differences among the scenarios, the mean percent deviations of the total emissions from the emission limit of all scenarios are compared as in Fig. 2. For each scenario, the mean deviations of the total emissions of all instances from their emission limits are found as percentages. Note that a positive deviation indicates that the emission is bought while a negative deviation indicates that the emission is sold. Because all scenarios

and their instances have different randomly generated parameters affecting the total emission (e.g., demand, capacity, and arc distance), the emission limit is determined proportionally to these parameters. Therefore, comparing the percent differences in total emission from the emission limit provides an unbiased comparison among all scenarios.

According to Fig. 2, the supply chain tends to buy emission allowances when the emission price is low, whereas it tends to sell emission allowances when the emission price is high for multimodal scenarios. The unimodal scenarios use significantly more emissions than the multimodal ones. The unimodal scenarios cannot reduce their total emissions by changing the transportation mode because they only have regular trucks. Therefore, the supply chain always buys emission allowances in the unimodal scenarios. However, the emission usage slightly reduces as the emission price increases since the model can change the manufacturing plants to reduce production emissions. Both unimodal and multimodal scenarios cause more emissions as the risk threshold decreases. The lower risk thresholds prevent the supply chain from using some unreliable but low emission suppliers whose transportation emissions are lower because of their closeness to the plants or access to rail or water transportation modes. Also, the total emission differences between two adjacent emission price levels may vary. For instance, the emission trade differences between the emission price levels of 0.3 and 0.6 are higher than the differences between the emission price levels of 0.9 and 1.2. The reason is that the difference between the marginal benefit of selling one unit of emission allowance (or buying one unit less emission) and the marginal cost of reducing one unit of emission changes at each emission price level. In other words, as the emission price increases, the reduction in carbon emissions becomes more attractive but the reduction becomes more challenging and costly. These trends are similar for all scenarios with different supply chain structures as seen in Fig. 2.

To analyze the emission variations of the instances within the scenarios, the percent deviations of the total emissions from the emission limits are compared in Figs. 3 and 4. Note that a positive deviation means that the total carbon emission exceeds the emission limit, while a negative deviation means that the total carbon emission is below the emission limit. The deviation becomes zero if the emission limit is fully used. Figures 3 and 4 show that the variation of instances within a scenario is not very high. Also, the total emission reduces as the risk threshold increases and the emission price increases.

As each scenario and its instances have various randomly generated parameters (e.g., demand, production cost, and arc distance) affecting total cost, the comparisons of a scenario with the others in terms of total cost become meaningless. Therefore, the percent deviation of the mean cost of a scenario from the mean of its supply chain structure is calculated to compare the variations in total cost among all scenarios. Note that the mean cost of a scenario is found by taking the average of 10 instances of a scenario (e.g., 2 suppliers, 2 plants, 2 customers, 3 products, 4 components, risk threshold level of 0.08, and emission price level of 0.3), whereas the mean cost of a supply chain structure (e.g., 2 suppliers, 2 plants, 2 customers, 3 products, 4 components) is found by taking the averages of all scenarios. Figure 5 presents the percentages of cost deviations for all scenarios. Note that a positive deviation means a higher cost while a negative deviation means a lower cost than the mean cost of the supply chain structure.

According to Fig. 5, the unimodal scenarios are more costly than the multimodal scenarios for all supply chain structures. Since the unimodal scenarios have only regular trucks and cannot reduce total emission by changing the transportation mode, total emissions of unimodal scenarios are always higher than the emission limit and emission allowances are always bought. Therefore, the total supply chain cost increases as emission price increases in unimodal scenarios. However, the trend of cost variations according to the emission price in

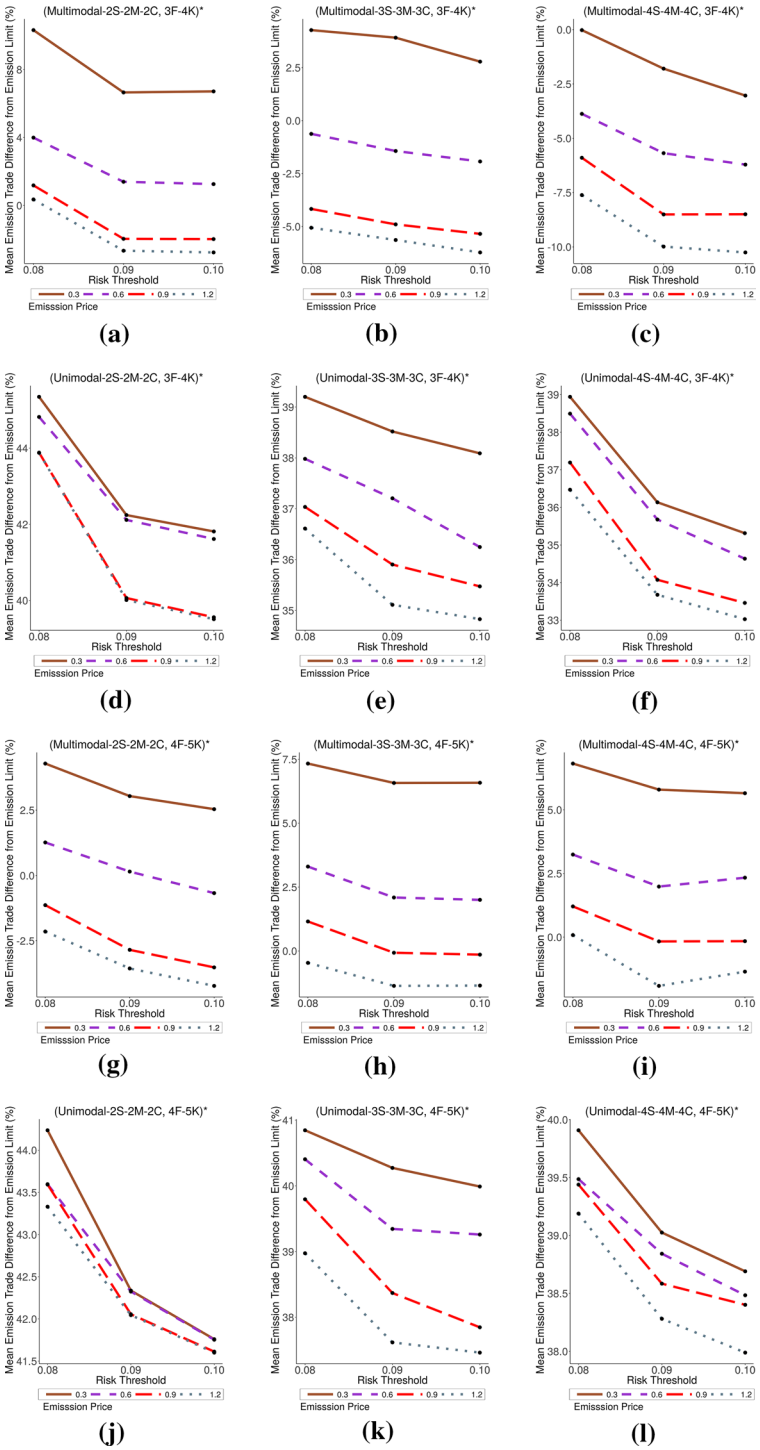


Fig. 2 Average percent deviations of total emission from the emission limit for all scenarios (*suppliers—manufacturing plants—customers, final products—components)

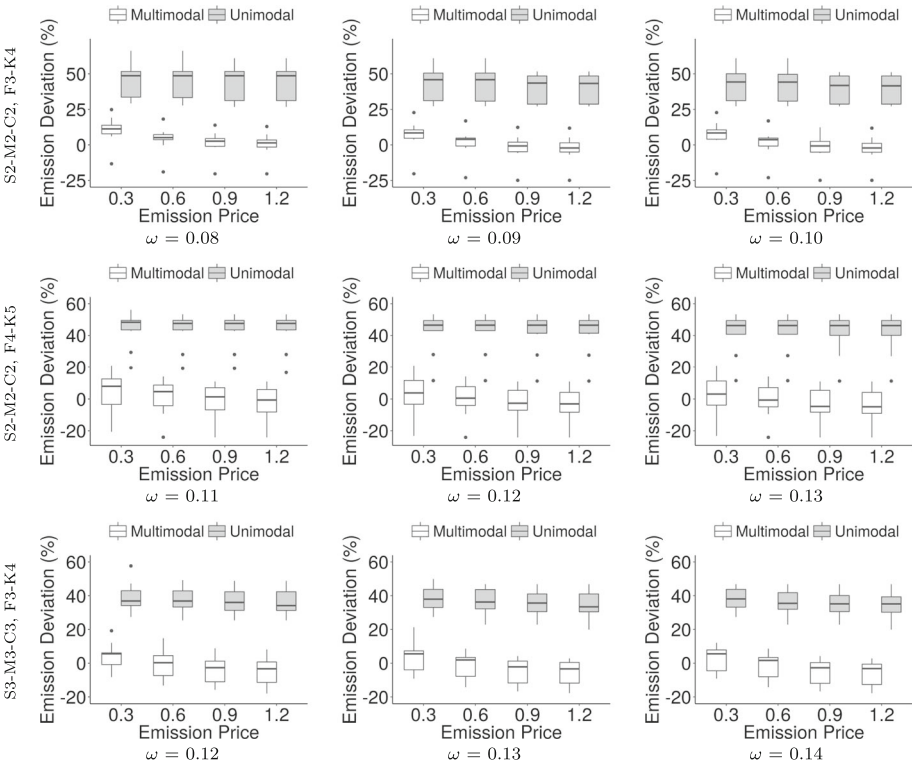


Fig. 3 Percent deviations of carbon emission from emission limit for 2S–2M–2C and 3S–3M–3C scenarios (*suppliers—manufacturing plants—customers, final products—components)

multimodal scenarios changes for different supply chain structures. In the multimodal scenarios, a break-even point between the marginal benefit of selling one unit emission allowance and the marginal cost of reducing one unit emission exists. When the emission price is above that point, the marginal benefit of selling emission is higher. When the emission price below that point, the marginal cost of reducing emissions is higher. Therefore, in some instances, total cost firstly increases as emission price increases up to the break-even point. After that point, the total cost decreases as emission price increases. For example, in the multimodal scenarios shown in Fig. 5a, b and g, the total cost increases as emission price increases up to the emission price level of 0.6 then it decreases as the emission price increases to 0.9 and 1.2. Similarly, the total cost increases as emission price increases up to the emission price level of 0.9 then it decreases as emission price increases to 1.2 in the multimodal scenarios shown in Fig. 5h and i. For the multimodal scenario shown in Fig. 5c, the total cost decreases as emission price increases because the marginal benefit of selling emission is higher for all emission price levels. On the other hand, the effect of the risk threshold on the total cost is the same for all scenarios. Total cost increases as the risk threshold decreases for all emission price levels of both multimodal and unimodal scenarios. Because, the company is enforced to buy components from more reliable but more costly suppliers as the risk threshold decreases.

Figures 6 and 7 show the average usage of transportation modes for all scenarios. Note that the transportation modes are the regular truck, environment-friendly truck, air, rail and water. The trucks (regular and environment-friendly) are the most used transportation options. Not

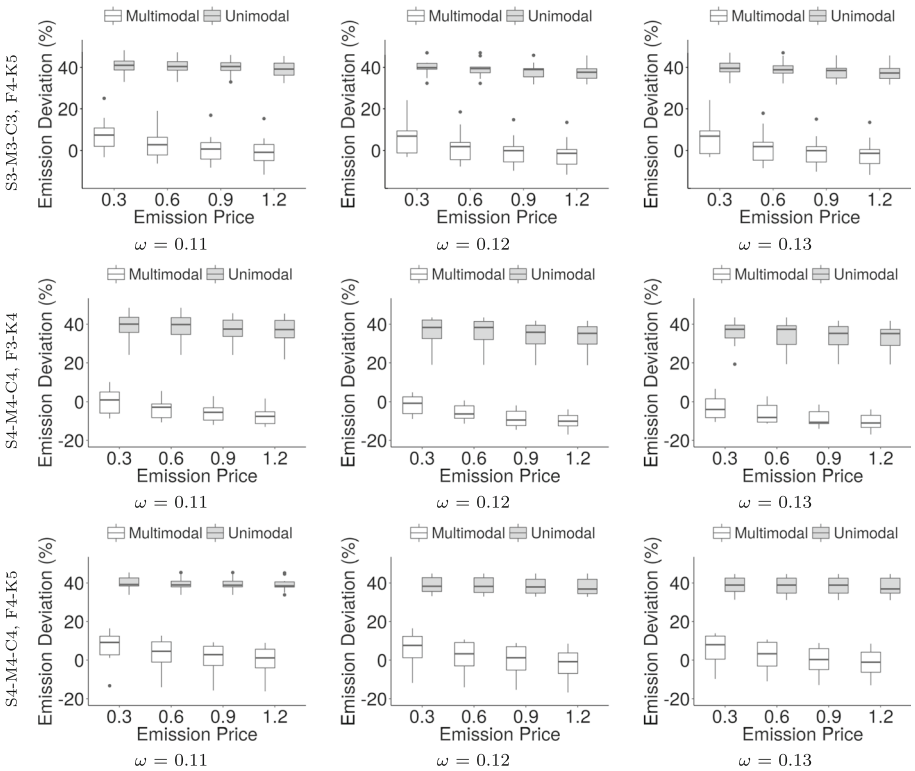


Fig. 4 Percent deviations of carbon emission from emission limit for 3S–3M–3C and 4S–4M–4C scenarios (*suppliers—manufacturing plants—customers, final products—components)

surprisingly, regular trucks are preferred with the low emission price scenarios, however, they are replaced by environment-friendly trucks in the high emission price scenarios. Rail has been significantly used, whereas water has been used as a small percentage due to its limited availability. Both rail and water slightly increase their share as the emission price increases. Not surprisingly, air transportation has not been used significantly due to its high cost and emissions. The risk threshold does not have a significant effect on the selection of transportation modes.

According to the results regarding the solution time, the multimodal scenarios require more time to solve than the unimodal scenarios. Furthermore, the scenarios with the lowest risk threshold level require more time to solve. However, this trend is not clearly visible when the number of nodes increases because most instances of large scenarios cannot be solved optimally within the time limit. The optimal solutions could not be found within the given time limit for some instances of the multimodal scenarios with supply chain structures of (3S–3M–3C, 4P–5K) and (4S–4M–4C, 3P–4K), and both multimodal and unimodal scenarios with the supply chain structure of (4S–4M–4C, 4P–5K). Note that the solution times are not reported due to the space limitations of this article.[1]

[1] All test problem data and their solutions are provided as an online supplement at https://okabadurmus.yasar.edu.tr/research/sc2018/.

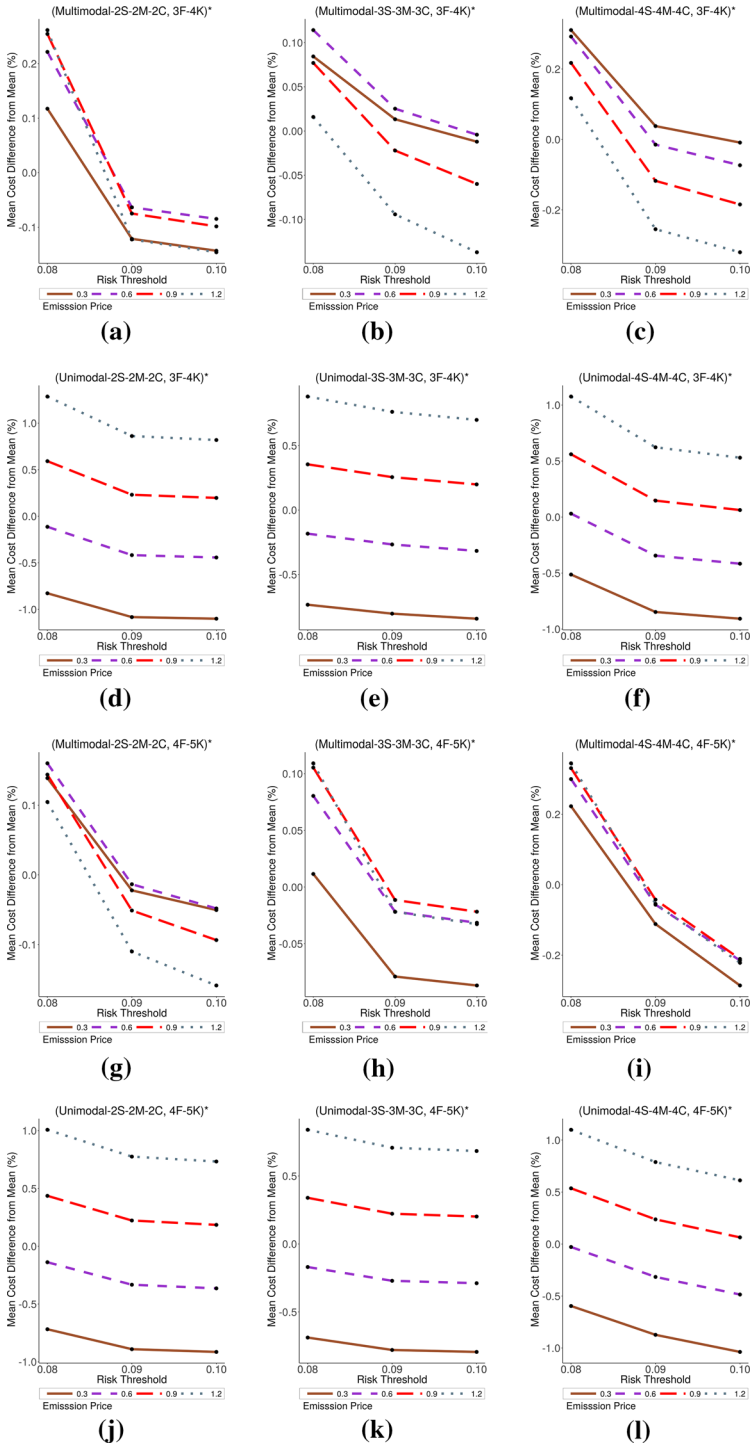


Fig. 5 Average percent deviations of total cost from the mean of the same supply chain structure for all scenarios (*suppliers—manufacturing plants—customers, final products—components)

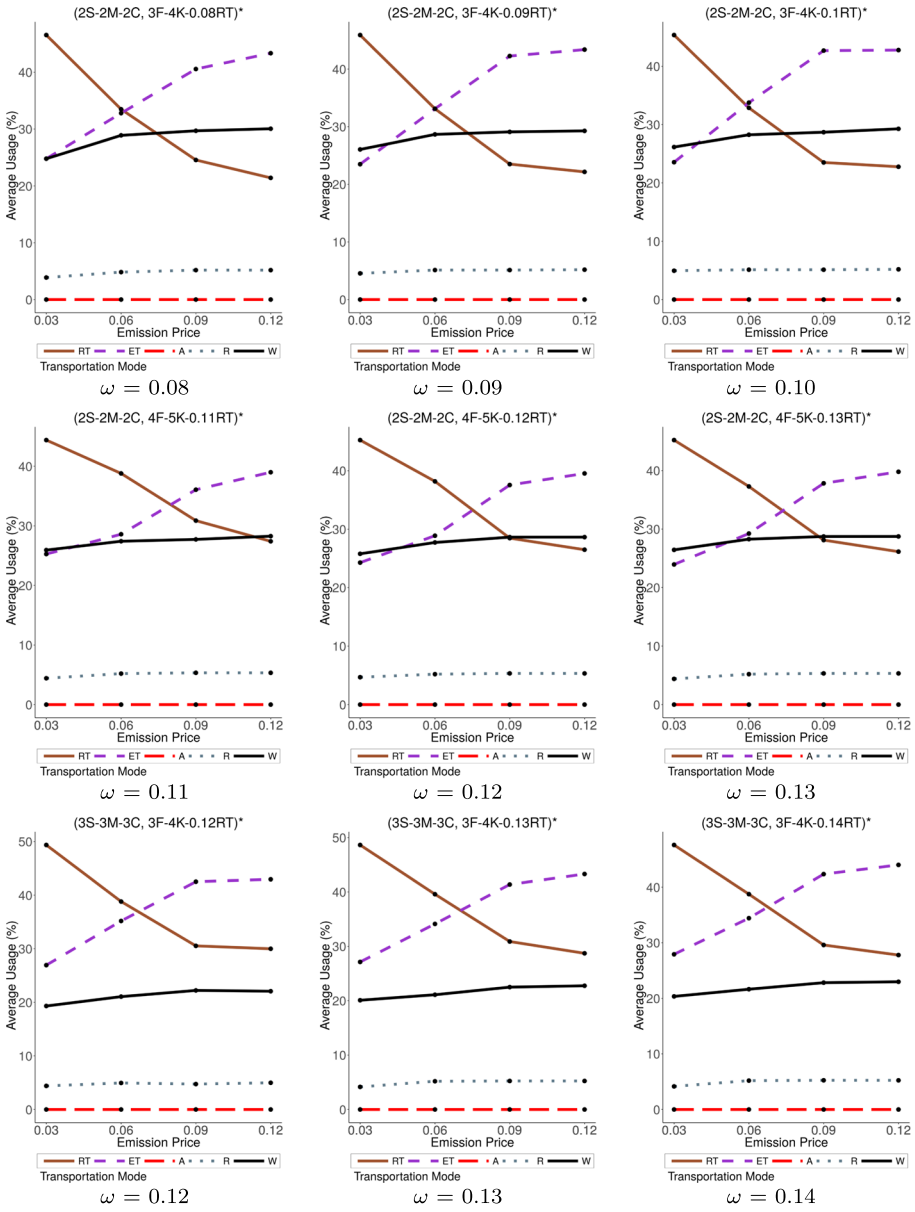


Fig. 6 Average usages (%) of transportation modes (*RT* regular truck, *ET* environment-friendly truck, *A* air, *R* rail, and *W* water) for 2S–2M–2C and 3S–3M–3C scenarios (*suppliers—manufacturing plants—customers, final products—components—risk threshold)

4.3 Discussion

This study focuses on supply risks as the recent studies (Jabbarzadeh et al. 2018; Fahimnia and Jabbarzadeh 2016) showed that supplier disruptions have a greater impact on supply chain cost than manufacturing disruptions. The results of this study showed that being risk-

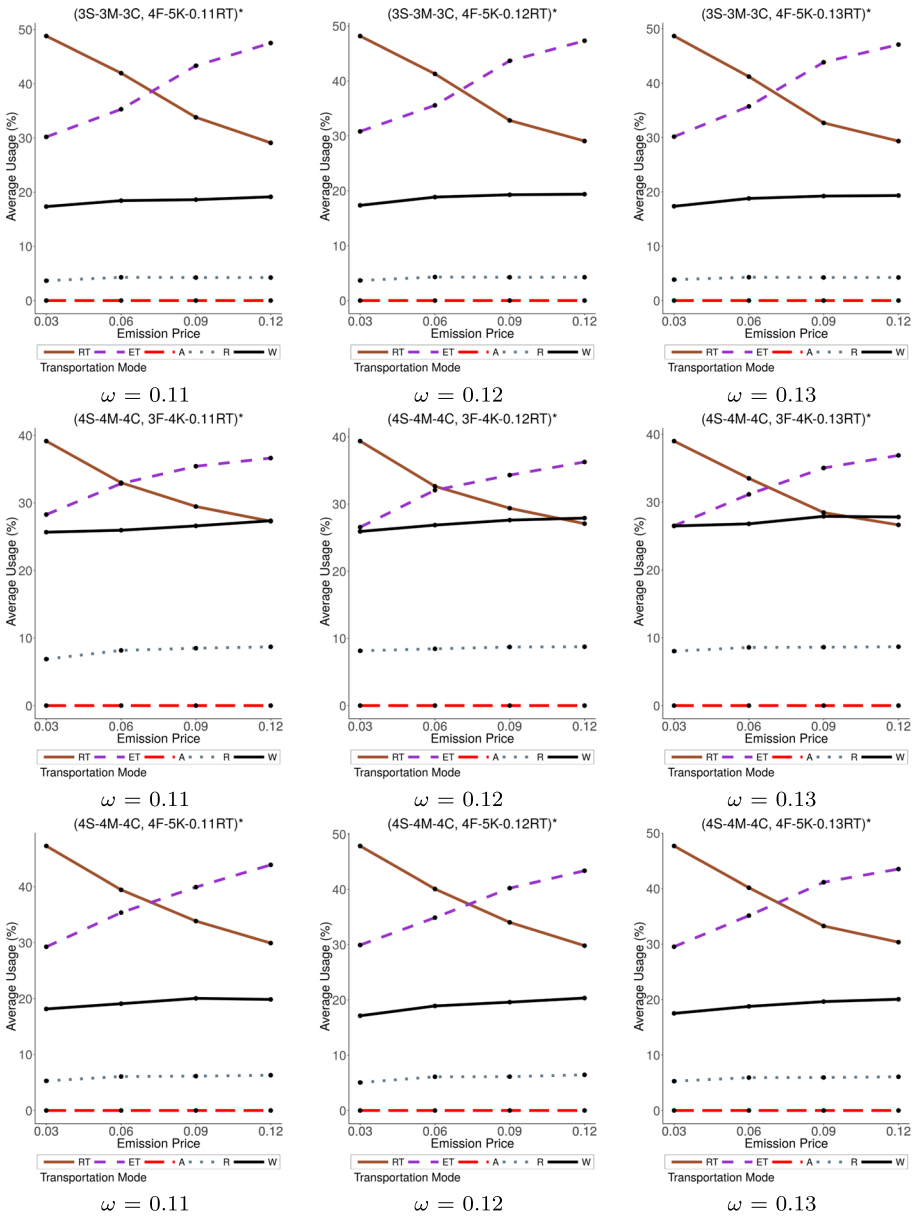


Fig. 7 Average usages (%) of transportation modes (*RT* regular truck, *ET* environment-friendly truck, *A* air, *R* rail, and *W* water) for 3S–3M–3C and 4S–4M–4C scenarios (*suppliers—manufacturing plants—customers, final products—components—risk threshold)

averse in supply operations increases the supply chain cost. And, decreasing risk below a certain threshold may be very costly. As presented in Fig. 5, the risk threshold can be relaxed to reduce total cost as the cost difference between the lowest and medium risk thresholds are higher (0.23% on average) than the difference (0.06% on average) between the medium and

the lowest risk thresholds. Therefore, the decision makers should select a risk threshold to balance supply chain costs and risks.

As mentioned in the literature review, far too little attention has been paid to multimodal transportation in supply chain network design literature. By comparing the unimodal and multimodal scenarios (Figs. 3 and 4), it can be observed that multimodality could benefit the distribution network in terms of emissions. The results showed that unimodal transportation causes 40.95% more emissions than multimodal transportation. Also, decision makers should utilize multimodal transportation to decrease the cost of the supply chain as multimodality reduces the cost by 1.20%. Figures 6 and 7 show that the usages of multimodal transport modes increase as emission price gets higher. These results are consistent with those of Liotta et al. (2015) and Shaw et al. (2016) who assessed the effect of multimodality on cost and emissions. Our results indicated that the usage of environment-friendly truck increases by 26.87%, 22.34% and 6.14% as emission price increases from 0.3 to 0.6, 0.6 to 0.9, and 0.9 to 1.2, respectively. Similarly, water and rail transportation modes are used more frequently as the emission price increases. However, increasing the emission price from 0.3 to 0.6 yields a more significant increase in the usage of multimodal transportation modes than increasing the emission price from 0.9 to 1.2. For example, the usage of water transportation increases by 20.49% and 3.52% as the emission price increases from 0.3 to 0.6 and from 0.9 to 1.2, respectively. A similar trend is observed in rail transportation mode, where the usage of rail transportation increases by 8.02% and 1.29% as emission price increases from 0.3 to 0.6 and from 0.9 to 1.2, respectively. In reviewing the literature, no data were found on the relationship between supply chain risk and transportation modes. In this study, as expected, the risk threshold does not appear to affect transportation mode decisions.

As for the sustainability aspect, the decision makers aiming to reduce carbon emissions of a supply chain should use multimodal transportation. Also, decreasing carbon emission of the supply chain operations and selling unused carbon allowance is only advisable when the emission price is high. Therefore, increasing the emission price is an effective method to reduce carbon emissions of companies as also stated by Waltho et al. (2019). For example, the emissions of multimodal scenarios are reduced by 3.55%, 2.65% and 1.16% as the emission price increases from 0.3 to 0.6, 0.6 to 0.9, and 0.9 to 1.2, respectively. This result shows that even a small increase in emission prices significantly reduces carbon emissions. Reducing carbon emissions can also be important if the carbon cap is reduced gradually per year. For example, the carbon cap in Ontario, Canada has been reduced by four percent per year from 2017 to 2020 (Environmental Commissioner of Ontario 2016). Furthermore, even if the environmentally harming operations have a small negative financial impact for low emission price levels, being environmentally conscious and reducing carbon emission will be beneficial for the reputation of the company as well as the welfare of the planet.

5 Conclusions

In this paper, a novel supply chain optimization problem is presented to design sustainable, reliable and multimodal supply chains. The objective function of our proposed model is to minimize the total supply chain cost including purchasing, production, supplier/plant activation and transportation costs. The presented model contributes to the literature in several ways. First, it includes the supply risk with a unique reliability formulation. Each supplier has a risk value and the model is constrained by a risk threshold which is determined according to the risk willingness of the decision maker. The proposed model guarantees a certain level

of total reliability using a risk threshold and the supply chain becomes more reliable as the risk threshold decreases. This type of risk modeling has not been addressed in the literature. Second, reliability, sustainability with a carbon cap-and-trade scheme, and multimodality are incorporated for the first time in the literature. Carbon emissions due to production and transportation operations are also included in the model. A carbon cap-and-trade scheme is considered and the remaining carbon emission under (over) the emission limit is sold (bought) at the emission market price. Third, the proposed model uniquely calculates the cost of air transportation according to the billable weight which simultaneously considers the physical weight and the volume of cargo.

The proposed model has been tested on a hypothetical but realistic case study. Results prove that if the decision maker is risk-averse, the total supply chain cost and carbon emissions are increased because risky but low cost (or low emission) suppliers cannot be chosen. As the risk threshold increases from the lowest to the highest, the total supply chain cost and emissions are increased by 0.29% and 1.99%, respectively. However, increasing the risk threshold from the lowest to medium level causes a more significant reduction in cost and emissions than increasing the risk threshold from medium to the highest. This suggests that setting the risk threshold at a medium level helps to balance supply risk with both cost and emission of a supply chain network. Although there was no study in the literature investigating the relationship between the supply risk and the use of multimodal transportation modes, our results show that the usage of transportation modes does not significantly change with different risk thresholds. Also, not surprisingly, multimodality reduces both supply chain cost and carbon emissions. The multimodal scenarios have 1.20% and 40.58% lower costs and emissions than unimodal scenarios, respectively. The results also show that the total emission is reduced in high emission price scenarios in order to reduce the total supply chain cost by selling the excess amount of emission limit. This confirms that the carbon cap-and-trade scheme is an effective tool to reduce carbon emissions. However, multimodal transportation is the main enabler to meet the cap-and-trade regulations, especially under tight emission limitations. This result should encourage companies to use multimodal transportation options more frequently. More importantly, the emission price has a positive effect to increase the use of more environment-friendly transportation options. Although the dominant transportation option of our study is road transportation, the use of rail and water options as well as environment-friendly trucks significantly increases as the emission price increases. However, the most significant increase in the use of the environment-friendly truck, water and rail transportation options are observed when the emission price is increased from the lowest to the medium levels. Therefore, decision makers can slightly increase the emission price to effectively increase the use of multimodal transportation options. On the other hand, the emissions can also be reduced by choosing the production plants with low emissions as observed in the unimodal scenarios where there is only one transportation mode is available. For example, in unimodal scenarios, the emissions are reduced by 1.90% as the emission price increases from the lowest level to the highest.

This study has several managerial implications. First, the proposed model can be used to determine the significant factors on the supply chain design and performance as it quantifies the supply risk, supply chain cost and total carbon emissions. Second, the results of the proposed model can be used by decision makers to choose an appropriate risk level to have a feasible and acceptable solution under certain budget considerations. Using the results of this study, the decision makers can choose the most appropriate suppliers for their supply operations in terms of cost and risk. Similarly, decision makers can select the best manufacturing sites to reduce costs and emissions. Third, the consideration of the supply chain cost, supply risk and the emissions of transportation and production operations in the proposed

model guarantees its applicability and feasibility to real-world applications, therefore, it can help managers to determine how they should invest in environment-friendly equipment with low emissions for strategically designing green supply chain networks.

This study has two limitations. First, the proposed model is tested on a hypothetical case study. Therefore, the general findings of the study may be limited. Application of the model to a real supply chain may reveal different relationships between sustainability, reliability and multimodality. Therefore, future studies should apply the model to a real-life case study to discover new findings or verify the current findings. Another direction for future study can be multi-objective modeling of the problem in order to analyze the trade-offs among cost, carbon emissions, and reliability. The second limitation is that the MILP model is effective up to 12 nodes in total. Thus, larger instances cannot be solved in a reasonable time due to the high complexity of the problem. This may be more important for real supply chains with large numbers of components and final products. To this end, future studies will focus on developing heuristic algorithms to solve the model within reasonable runtimes.

Acknowledgements This research did not receive any specific Grant from funding agencies in the public, commercial, or not-for-profit sectors.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

Ahranjani, P. M., Ghaderi, S. F., Azadeh, A., & Babazadeh, R. (2020). Robust design of a sustainable and resilient bioethanol supply chain under operational and disruption risks. *Clean Technologies and Environmental Policy*, *22*(1), 119–151.

Alizadeh, M., Ma, J., Marufuzzaman, M., & Yu, F. (2019). Sustainable olefin supply chain network design under seasonal feedstock supplies and uncertain carbon tax rate. *Journal of Cleaner Production*, *222*, 280–299.

Avi-Yonah, R. S., & Uhlmann, D. M. (2009). Combating global climate change: Why a carbon tax is a better response to global warming than cap and trade. *Stanford Environmental Law Journal*, *28*, 3–50.

Chaabane, A., Ramudhin, A., & Paquet, M. (2011). Designing supply chains with sustainability considerations. *Production Planning & Control*, *22*(8), 727–741. https://doi.org/10.1080/09537287.2010.543554.

Chopra, S., & Sodhi, M. S. (2004). Managing risk to avoid supply-chain breakdown. *MIT Sloan Management Review*, *46*, 53–62.

Choudhary, A., Sarkar, S., Settur, S., & Tiwari, M. (2015). A carbon market sensitive optimization model for integrated forward-reverse logistics. *International Journal of Production Economics*, *164*, 433–444. https://doi.org/10.1016/j.ijpe.2014.08.015.

Diabat, A., & Al-Salem, M. (2015). An integrated supply chain problem with environmental considerations. *International Journal of Production Economics*, *164*, 330–338. https://doi.org/10.1016/j.ijpe.2014.12.004.

Environmental Commissioner of Ontario. (2016). Introduction to cap and trade in ontario: Appendix a to the ecos greenhouse gas progress report 2016. Progress report. Technical report, Environmental Commissioner of Ontario. https://media.assets.eco.on.ca/web/2016/11/Appendix-A-Introduction-to-Cap-and-Trade-in-Ontario.pdf. Retrieved 15 February, 2020.

Fahimnia, B., & Jabbarzadeh, A. (2016). Marrying supply chain sustainability and resilience: A match made in heaven. *Transportation Research Part E: Logistics and Transportation Review*, *91*, 306–324.

Fahimnia, B., Jabbarzadeh, A., & Sarkis, J. (2018). Greening versus resilience: A supply chain design perspective. *Transportation Research Part E: Logistics and Transportation Review*, *119*, 129–148.

Fahimnia, B., Sarkis, J., Boland, J., Reisi, M., & Goh, M. (2015). Policy insights from a green supply chain optimisation model. *International Journal of Production Research*, *53*(21), 6522–6533. https://doi.org/10.1080/00207543.2014.958592.

Goulder, L. H., & Schein, A. R. (2013). Carbon taxes versus cap and trade: A critical review. *Climate Change Economics*, *4*(3), 1350010–1–1350010–28.

Han, S., Jiang, Y., Zhao, L., Leung, S. C., & Luo, Z. (2017). Weight reduction technology and supply chain network design under carbon emission restriction. *Annals of Operations Research*,. https://doi.org/10.1007/s10479-017-2696-8.

Hosseini, S., Ivanov, D., & Dolgui, A. (2019). Review of quantitative methods for supply chain resilience analysis. *Transportation Research Part E: Logistics and Transportation Review*, *125*, 285–307. https://doi.org/10.1016/j.tre.2019.03.001.

Ho, W., Zheng, T., Yildiz, H., & Talluri, S. (2015). Supply chain risk management: A literature review. *International Journal of Production Research*, *53*(16), 5031–5069. https://doi.org/10.1080/00207543.2015.1030467.

Jabbarzadeh, A., Fahimnia, B., & Sabouhi, F. (2018). Resilient and sustainable supply chain design: Sustainability analysis under disruption risks. *International Journal of Production Research*, *56*(17), 5945–5968.

Kaur, H., & Singh, S. P. (2019). Sustainable procurement and logistics for disaster resilient supply chain. *Annals of Operations Research*, *283*(1), 309–354.

Kull, T. J., & Talluri, S. (2008). A supply risk reduction model using integrated multicriteria decision making. *IEEE Transactions on Engineering Management*, *55*(3), 409–419. https://doi.org/10.1109/TEM.2008.922627.

Le, T. P. N., & Lee, T. R. (2013). Model selection with considering the CO_2 emission alone the global supply chain. *Journal of Intelligent Manufacturing*, *24*(4), 653–672. https://doi.org/10.1007/s10845-011-0613-6.

Liotta, G., Stecca, G., & Kaihara, T. (2015). Optimisation of freight flows and sourcing in sustainable production and transportation networks. *International Journal of Production Economics*, *164*, 351–365. https://doi.org/10.1016/j.ijpe.2014.12.016.

McKinnon, A., & Piecyk, M. (2010). Measuring and managing CO^2 emissions. Technical report, European Chemical Industry Council (CEFIC). https://cefic.org/app/uploads/2018/12/MeasuringAndManagingCO2EmissionOfEuropeanTransport-McKinnon-24.01.2011-REPORT_TRANSPORT_AND_LOGISTICS.pdf. Retrieved 3 February, 2020.

Moradinasab, N., Amin-Naseri, M., Behbahani, T. J., & Jafarzadeh, H. (2018). Competition and cooperation between supply chains in multi-objective petroleum green supply chain: A game theoretic approach. *Journal of Cleaner Production*, *170*, 818–841. https://doi.org/10.1016/j.jclepro.2017.08.114.

Piecyk, M., Browne, M., Whiteing, A., & McKinnon, A. (2015). *Green logistics: Improving the environmental sustainability of logistics*. London: Kogan Page Publishers.

Rabbani, M., Saravi, N. A., Farrokhi-Asl, H., Lim, S. F. W., & Tahaei, Z. (2018). Developing a sustainable supply chain optimization model for switchgrass-based bioenergy production: A case study. *Journal of Cleaner Production*, *200*, 827–843. https://doi.org/10.1016/j.jclepro.2018.07.226.

Rahmani, D., & Mahoodian, V. (2017). Strategic and operational supply chain network design to reduce carbon emission considering reliability and robustness. *Journal of Cleaner Production*, *149*, 607–620.

Ramudhin, A., Chaabane, A., Kharoune, M., & Paquet, M. (2008). Carbon market sensitive green supply chain network design. In: *2008 IEEE international conference on industrial engineering and engineering management* (pp. 1093–1097). https://doi.org/10.1109/IEEM.2008.4738039.

Rezaee, A., Dehghanian, F., Fahimnia, B., & Beamon, B. (2015). Green supply chain network design with stochastic demand and carbon price. *Annals of Operations Research*,. https://doi.org/10.1007/s10479-015-1936-z.

Rezaee, A., Dehghanian, F., Fahimnia, B., & Beamon, B. (2017). Green supply chain network design with stochastic demand and carbon price. *Annals of Operations Research*, *250*(2), 463–485.

Shaw, K., Irfan, M., Shankar, R., & Yadav, S. S. (2016). Low carbon chance constrained supply chain network design problem: A benders decomposition based approach. *Computers & Industrial Engineering*, *98*, 483–497.

Shaw, K., Shankar, R., Yadav, S. S., & Thakur, L. S. (2013). Modeling a low-carbon garment supply chain. *Production Planning & Control*, *24*(8–9), 851–865.

Trappey, A. J. C., Peng, T. K., & Lin, H. D. (1996). An object-oriented bill of materials system for dynamic product management. *Journal of Intelligent Manufacturing*, *7*(5), 365–371. https://doi.org/10.1007/BF00123912.

Tummala, R., & Schoenherr, T. (2011). Assessing and managing risks using the supply chain risk management process (SCRMP). *Supply Chain Management: An International Journal*, *16*(6), 474–483. https://doi.org/10.1108/13598541111171165.

Tuncel, G., & Alpan, G. (2010). Risk assessment and management for supply chain networks: A case study. *Computers in Industry*, *61*(3), 250–259. https://doi.org/10.1016/j.compind.2009.09.008.

Waltho, C., Elhedhli, S., & Gzara, F. (2019). Green supply chain network design: A review focused on policy adoption and emission quantification. *International Journal of Production Economics*, *208*, 305–318. https://doi.org/10.1016/j.ijpe.2018.12.003.

World Bank Group. (2017). Carbon pricing watch 2017. Technical report, World Bank. http://documents.worldbank.org/curated/en/699641497346643090/Carbon-pricing-watch-2017. Retrieved 6 February, 2020.

Wu, T., Blackhurst, J., & Chidambaram, V. (2006). A model for inbound supply risk analysis. *Computers in Industry*, *57*(4), 350–365. https://doi.org/10.1016/j.compind.2005.11.001.

Zakeri, A., Dehghanian, F., Fahimnia, B., & Sarkis, J. (2015). Carbon pricing versus emissions trading: A supply chain planning perspective. *International Journal of Production Economics*, *164*, 197–205. https://doi.org/10.1016/j.ijpe.2014.11.012.

Zhang, D., Zou, F., Li, S., & Zhou, L. (2017). Green supply chain network design with economies of scale and environmental concerns. *Journal of Advanced Transportation*,. https://doi.org/10.1155/2017/6350562.

Zissis, D., Saharidis, G. K., Aktas, E., & Ioannou, G. (2018). Emission reduction via supply chain coordination. *Transportation Research Part D: Transport and Environment*, *62*, 36–46.

Zsidisin, G. A., & Ellram, L. M. (2003). An agency theory investigation of supply risk m anagement. *Journal of Supply Chain Management*, *39*(2), 15–27. https://doi.org/10.1111/j.1745-493X.2003.tb00156.x.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.